

Unit 11: Quadratic Formula & Applications

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Session 11.1:

Quadratic Formula, Discriminant, Number of Solutions

The Quadratic Formula

For any quadratic equation in standard form $ax^2 + bx + c = 0$ where $a \neq 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

This formula gives you the solutions (roots) of the equation.

Using the Quadratic Formula

Steps:

1. Write the equation in standard form: $ax^2 + bx + c = 0$
2. Identify a , b , and c
3. Substitute into the formula
4. Simplify under the square root first
5. Simplify the entire expression
6. Write two solutions using the $\pm$ symbol

The Discriminant

The discriminant is the expression under the square root: $b^2 - 4ac$

The discriminant tells you how many real solutions the equation has:

If $b^2 - 4ac > 0$: Two different real solutions

If $b^2 - 4ac = 0$: One real solution (a repeated root)

If $b^2 - 4ac < 0$: No real solutions (two complex solutions)

Simplifying Square Roots

Remember to simplify radicals in your final answer.

If you have a common factor in the numerator, factor it out.

Homework 11.1:

Quadratic Formula, Discriminant, Number of Solutions

Part A: Identify a , b , and c

For each equation, identify the values of a , b , and c .

Problem 1:

$$x^2 + 5x + 6 = 0$$

$$a = \underline{\hspace{2cm}} \quad b = \underline{\hspace{2cm}} \quad c = \underline{\hspace{2cm}}$$

Problem 2:

$$3x^2 - 7x + 2 = 0$$

$$a = \underline{\hspace{2cm}} \quad b = \underline{\hspace{2cm}} \quad c = \underline{\hspace{2cm}}$$

Problem 3:

$$2x^2 + 8x = 10$$

First rewrite in standard form: $\underline{\hspace{10cm}}$

$$a = \underline{\hspace{2cm}} \quad b = \underline{\hspace{2cm}} \quad c = \underline{\hspace{2cm}}$$

Part B: Use the Discriminant

Find the discriminant and determine the number of real solutions.

Problem 4:

$$x^2 + 6x + 9 = 0$$

Discriminant: _____

Number of real solutions: _____

Problem 5:

$$x^2 + 6x - 9 = 0$$

Discriminant: _____

Number of real solutions: _____

Problem 6:

$$2x^2 + 6x + 9 = 0$$

Discriminant: _____

Number of real solutions: _____

Part C: Solve Using the Quadratic Formula

Solve each equation using the quadratic formula. Show all work.

Give exact answers (simplified radicals when needed).

Problem 7:

$$x^2 + 3x - 4 = 0$$

Problem 8:

$$x^2 - 8x + 16 = 0$$

Problem 9:

$$2x^2 + 8x + 9 = 0$$

Part D: Equations That Need Rearranging

First rewrite in standard form, then solve using the quadratic formula.

Problem 10:

$$-x^2 = 4x - 6$$

Problem 11:

$$2x^2 + 8 = 10x$$

Part E: Challenge Problems

Problem 12:

Find a value of c so that $x^2 + 12x + c = 0$ has exactly one real solution.

Session 11.2:

Applications: Projectile/Area Models

Projectile Motion

When an object is thrown or launched, its height h (in feet) after t seconds is modeled by:

$$h = -16t^2 + v_0t + h_0$$

where:

- -16 is half the acceleration due to gravity (in ft/s^2)
- v_0 is the initial velocity (in ft/s)
- h_0 is the initial height (in feet)

Important questions:

- When does the object hit the ground? (Set $h = 0$)
- What is the maximum height? (Find the vertex)
- When does it reach a certain height? (Set $h = \text{that height}$)

Area Problems

Many geometry problems involve quadratic equations:

- Rectangle: $\text{Area} = \text{length} \times \text{width}$
- Square: $\text{Area} = \text{side}^2$
- Triangle: $\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$

Checking Reasonableness

Always check if your answer makes sense in context:

- Time cannot be negative (usually)
- Lengths and distances must be positive
- Check units match the problem
- Does the magnitude make sense?

Homework 11.2:

Applications: Projectile/Area Models

Part A: Projectile Motion

Problem 1:

A ball is thrown upward from ground level with an initial velocity of 48 feet per second.

Its height h in feet after t seconds is given by:

$$h = -16t^2 + 48t$$

Part A:

When does the ball hit the ground? Show your work.

Part B:

What is the maximum height of the ball?

Part C:

At what time(s) is the ball at a height of 32 feet?

Problem 2:

A rocket is launched from a platform 10 feet above the ground.

Its initial upward velocity is 80 feet per second.

Its height h in feet after t seconds is given by:

$$h = -16t^2 + 80t + 10$$

Part A:

What is the initial height of the rocket?

Part B:

Find the maximum height of the rocket and when it occurs.

Part C:

When does the rocket hit the ground? (Round to 2 decimal places)

Problem 3:

An arrow is shot upward from a height of 5 feet.

Its height h in feet after t seconds is given by:

$$h = -16t^2 + 64t + 5$$

Part A:

At what time(s) is the arrow at a height of 53 feet?

Part B:

When does the arrow return to its initial height of 5 feet?

Part C:

Explain why one of these times is $t = 0$. What does the other time represent?

Part B: Area Problems

Problem 4:

A rectangle has a length that is 4 inches more than its width.

The area of the rectangle is 60 square inches.

Part A:

Let w represent the width in inches. Write an equation for the area.

Part B:

Solve for w using the quadratic formula.

Part C:

Which solution is reasonable? What is the length of the rectangle?

Problem 5:

A square picture has a frame around it.

The frame is 2 inches wide on all sides.

The total area (picture plus frame) is 120 square inches.

Part A:

Let s represent the side length of the picture in inches.

Write an expression for the side length of the entire square (picture + frame).

Part B:

Write an equation for the total area.

Part C:

Solve for s . What is the side length of the picture?

Part D:

Check: Does your answer make sense? Verify the total area is 120 square inches.

Problem 6:

A rectangular garden has a length that is 3 meters more than twice its width.

The area of the garden is 35 square meters.

Part A:

Let w represent the width in meters. Write an expression for the length.

Part B:

Write an equation for the area.

Part C:

Solve for w . What are the dimensions of the garden?

Part D:

Explain why you reject one of the solutions.

Part C: Mixed Applications

Problem 7:

A baseball is hit from a height of 3 feet.

Its height h in feet after t seconds is given by:

$$h = -16t^2 + 40t + 3$$

The outfield fence is 10 feet tall.

Will the ball clear the fence if it reaches the fence at $t = 2$ seconds?

Show your work.

Problem 8:

A triangle has a base of x centimeters and a height of $(x + 5)$ centimeters.

The area of the triangle is 42 square centimeters.

Part A:

Write an equation for the area.

Part B:

Solve for x . What are the base and height of the triangle?

Session 11.3:

Mixed Methods: Choose Factor, Square Root, Formula

Three Methods for Solving Quadratics

Method 1: Factoring

Best when: The quadratic factors easily

Pros: Fast and simple when it works

Cons: Doesn't always work; can be hard to see factors

Method 2: Square Roots

Best when: Equation is in the form $(x - h)^2 = k$ or $ax^2 + c = 0$

Pros: Very fast for these special forms

Cons: Only works for specific types of equations

Method 3: Quadratic Formula

Best when: Factoring is difficult or impossible

Formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Pros: Always works for any quadratic

Cons: More steps; arithmetic can be messy

Decision Guide

If it is in the form $ax^2 + c = 0$ or $(x - h)^2 = k \rightarrow$ Use square roots

If you can easily see factors $\rightarrow$ Use factoring

If factoring is hard or you're unsure $\rightarrow$ Use the quadratic formula

Homework 11.3:

Mixed Methods: Choose Factor, Square Root, Formula

Part A: Identify the Best Method

For each equation, identify which method would be most efficient. Do not solve.

Problem 1:

$$x^2 - 64 = 0$$

Best method: _____

Why? _____

Problem 2:

$$x^2 + 7x + 12 = 0$$

Best method: _____

Why? _____

Problem 3:

$$2x^2 + 5x - 3 = 0$$

Best method: _____

Why? _____

Problem 4:

$$(x + 4)^2 = 25$$

Best method: _____

Why? _____

Problem 5:

$$3x^2 - 27 = 0$$

Best method: _____

Why? _____

Problem 6:

$$x^2 + 6x + 2 = 0$$

Best method: _____

Why? _____

Part B: Solve Using the Best Method

For each equation, choose the most efficient method and solve. State which method you used.

Problem 7:

$$x^2 - 9 = 0$$

Method used: _____

Problem 8:

$$x^2 - 8x + 15 = 0$$

Method used: _____

Problem 9:

$$x^2 + 4x - 1 = 0$$

Method used: _____

Problem 10:

$$4x^2 = 100$$

Method used: _____

Problem 11:

$$2x^2 - 5x - 3 = 0$$

Method used: _____

Problem 12:

$$(x - 7)^2 = 12$$

Method used: _____

Part C: More Practice—Your Choice of Method

Solve each equation using any method you choose.

Show your work and state your method.

Problem 13:

$$x^2 - 10x + 21 = 0$$

Problem 14:

$$3x^2 + 7x + 2 = 0$$

Problem 15:

$$5x^2 - 80 = 0$$

Problem 16:

$$x^2 + 2x - 5 = 0$$

Problem 17:

$$(x + 3)^2 - 16 = 0$$

Problem 18:

$$x^2 - 12x + 36 = 0$$

Part D: Application Problems

Solve using any method. Show your work.

Problem 19:

A ball is thrown upward from ground level.

Its height h in feet after t seconds is:

$$h = -16t^2 + 48t$$

When does the ball hit the ground?

Which method did you use and why?

Problem 20:

A rectangular garden has a width of x meters.

Its length is 7 meters more than its width.

The area is 120 square meters.

Find the dimensions of the garden.

Unit 11 Quiz:

Quadratic Formula & Applications

Section 1: The Discriminant

Problem 1:

Find the discriminant of $x^2 - 6x + 9 = 0$.

How many real solutions does the equation have?

Problem 2:

Find the discriminant of $2x^2 + 5x + 7 = 0$.

How many real solutions does the equation have?

Section 2: Solve Using the Quadratic Formula

Problem 3:

Solve $x^2 + 8x + 12 = 0$ using the quadratic formula.

Problem 4:

Solve $2x^2 - 6x + 1 = 0$ using the quadratic formula.

Simplify your answer completely.

Section 3: Applications

Problem 5:

A stone is thrown upward from a cliff 32 feet above the ground.

Its initial upward velocity is 64 feet per second.

Its height h in feet after t seconds is:

$$h = -16t^2 + 64t + 32$$

Part A:

What is the maximum height of the stone?

Part B:

When does the stone hit the ground? (Round to 2 decimal places)

Problem 6:

A rectangle has a width of w inches.

Its length is 5 inches more than twice its width.

The area of the rectangle is 75 square inches.

Part A:

Write an equation for the area.

Part B:

Solve for w and find the dimensions of the rectangle.

Section 4: Choose Your Method

Solve each equation using the most efficient method. State which method you used.

Problem 7:

$$x^2 - 16 = 0$$

Method: _____

Problem 8:

$$x^2 + 9x + 20 = 0$$

Method: _____

Problem 9:

$$x^2 + 6x + 4 = 0$$

Method: _____